MIL-P-13800 (Ord)
22 November 1954
SUPERSEDING
PA-PD-362
7 July 1953

MILITARY SPECIFICATION

PERLITE

1. SCOPE

1.1 This specification covers perlite which is intended for use as an ingredient in fillers for inert ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, and all specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard, For Testing Purposes

MILITARY

MIL-G-2550 -General Specification for Ammunition Except Small Arms Ammunition
MIL-P-10025 -Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification For

(Copies of specifications, standards, drawings, and publications, required by contractors in connections with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- The material shall consist of an expanded volcanic ash composed principally of silicon dioxide and aluminum trioxide.

3.2 Granulation.-Perlite shall conform to the granulation requirements specified in Table I using U. S. Standard Sieves conforming to Specification RR-S-366 when tested as specified in 4.4.

FSC 1395

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TABLE I .- Granulation.

Thru Sieve No.	Percentage
30	90, minimum
50	40 to 75
100	10 to 40
200	25, maximum

3.3 Apparent density.-0.10 to 0.30 gm per milliliter, when tested as specified in 4.5.

3.4 Crushing strength.-75 psi, minimum, when tested as specified in 4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of material from not more than one manufacturing batch of perlite. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

4.2 Sampling.-Ten percent, but in no case more than 10 or fewer than 3, of the containers in the lot shall be selected by the inspector so as to be representative of the lot. If there are fewer than three containers in the lot, all the containers shall be sampled. The material shall be mixed to a uniform consistency throughout and approximately 1 pound of it shall be taken from each selected container. Each of these primary samples shall be placed in a clean, moisture-free, air-tight container, sealed and labeled so that the container, from which it has been taken can be identified. A composite sample of approximately one pound shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in a clean, moisture-free, air-tight container, sealed and labeled to show the name of the material, manufacturer, plant, contract or order number, lot number and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with the requirements of this specification. All primary samples shall be held for possible future examination should the composite sample fail to comply with the requirements.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.4 Test for granulation.- A next of the prescribed U. S. Standard Sieves, arranged from top to bottom in order of increasing sieve numbers, shall be prepared. A bottom pan shall be placed under the lower sieve and 100gm of the sample shall be transferred to the upper sieve. The upper sieve shall be covered and shaken for 3 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. The portions retained by the various sieves shall be weighed. The percentages of perlite passing through the sieves shall be calculated.

4.5 Test for apparent density.- A funnel having a stem of 1/4 inch internal diameter shall be placed so that the tip of the stem shall be just above the uppermost graduation of a tared 100 ml graduated cylinder. The cylinder shall be filled to the 100 ml mark by pouring small quantities of the sample on the side of the funnel, allowing it to fall by gravity through the stem. Avoid jarring or tapping the cylinder. The cylinder shall be weighed. The gain in weight divided by 100 shall be reported as the apparent density.

4.6 Test for crushing strength. -

4.6.1 Apparatus.-The apparatus (see figure 1), for measuring the crushing

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strength of perlite shall consist of a steel cylinder, a steel cylinder base, a steel piston fitting snugly inside the cylinder, a steel guide collar to hold the piston in line with the cylinder, and a press having a capacity of at least 2 tons and a travel rate of 120 / 10 inches per minute.

4.6.2 Procedure.-The cylinder shall be on its base and filled with perlite to within 2 inches of the top. The cylinder shall be bounced on a firm surface 25 times to consolidate the sample. The guide collar shall be placed on the top of the cylinder and the cylinder and guide collar filled to within $\frac{1}{2}$ inch of the top of the guide collar. The cylinder shall be bounced an additional 25 times to consolidate the perlite. The guide collar shall be removed,and the perlite smoothed off flush with the top of the cylinder. Replace the collar on the cylinder, and compress the perlite to a depth of 2 inches below the top of the cylinder, using the piston and the press. Note the load and report the load divided by the cross sectional area of the piston as the crushing strength of the sample.

4.7 Resubmission and retest.-If the composite sample, or any primary sample subjected to test, fails to pass the test, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples from any or all of the containers in the lot at no expense to the government. The contractor may then remove the defective portions of the lot and resubmit the lot for acceptance. The resubmitted lot shall be accepted provided that new samples, selected in accordance with paragraph 4.2, pass all the tests required by this specification and the 3 additional primary samples pass any test or tests failed by the material on original submission.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling and marking.-Unless otherwise specified, each package of perlite shall be legibly marked with a description of the contents, the name of the manufacturer, the brand, (if any), the net weight of the contents, the name of the contractor,and the number of the contract or purchase order. In addition, packaging, packing, labeling,and marking shall be as specified in Specification MIL-P-10025.

6. NOTES

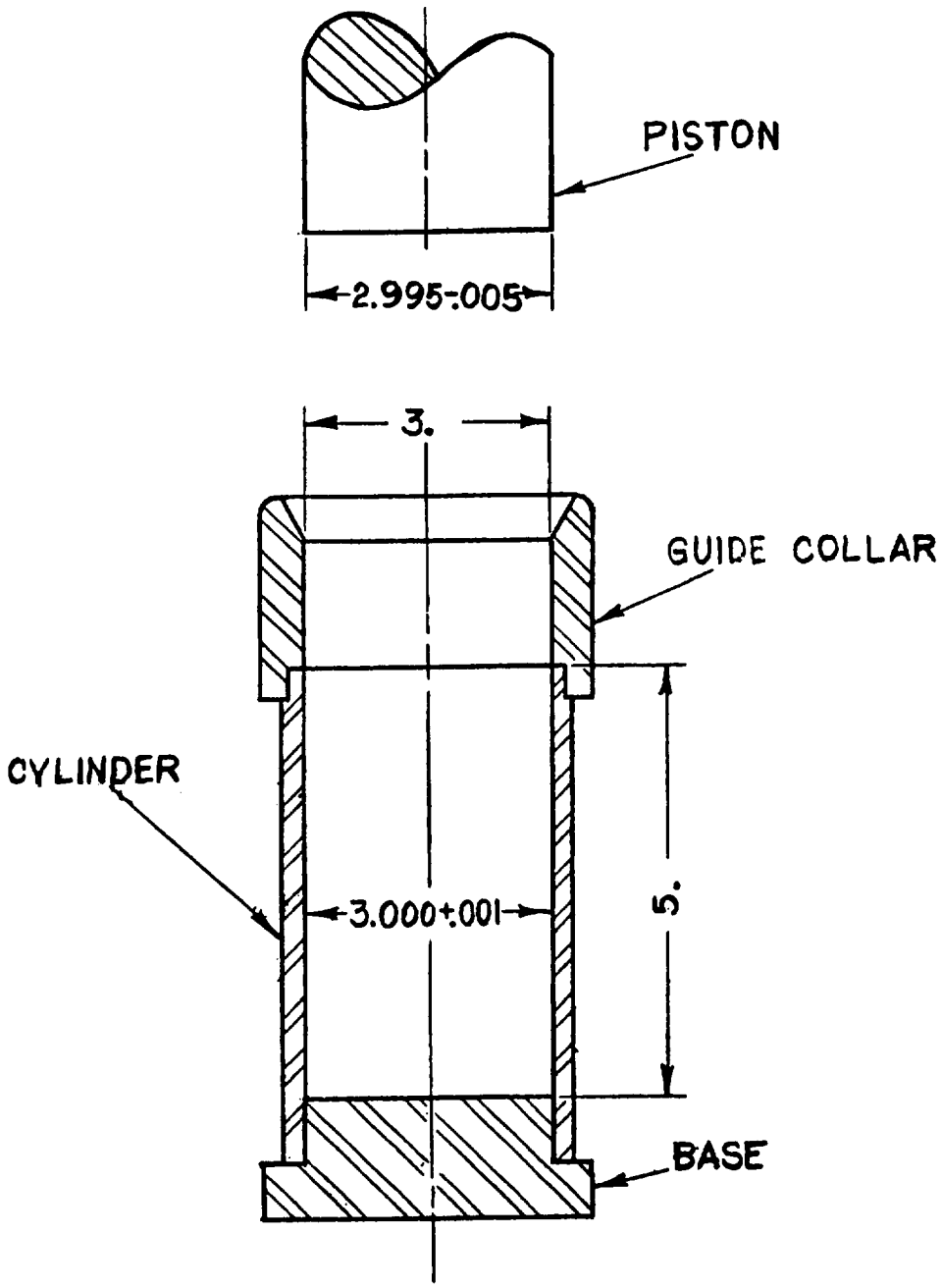
6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligations whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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Army- MU

Preparing Activity: Army- MU

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CRUSHING STRENGTH TEST APPARATUS

FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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